

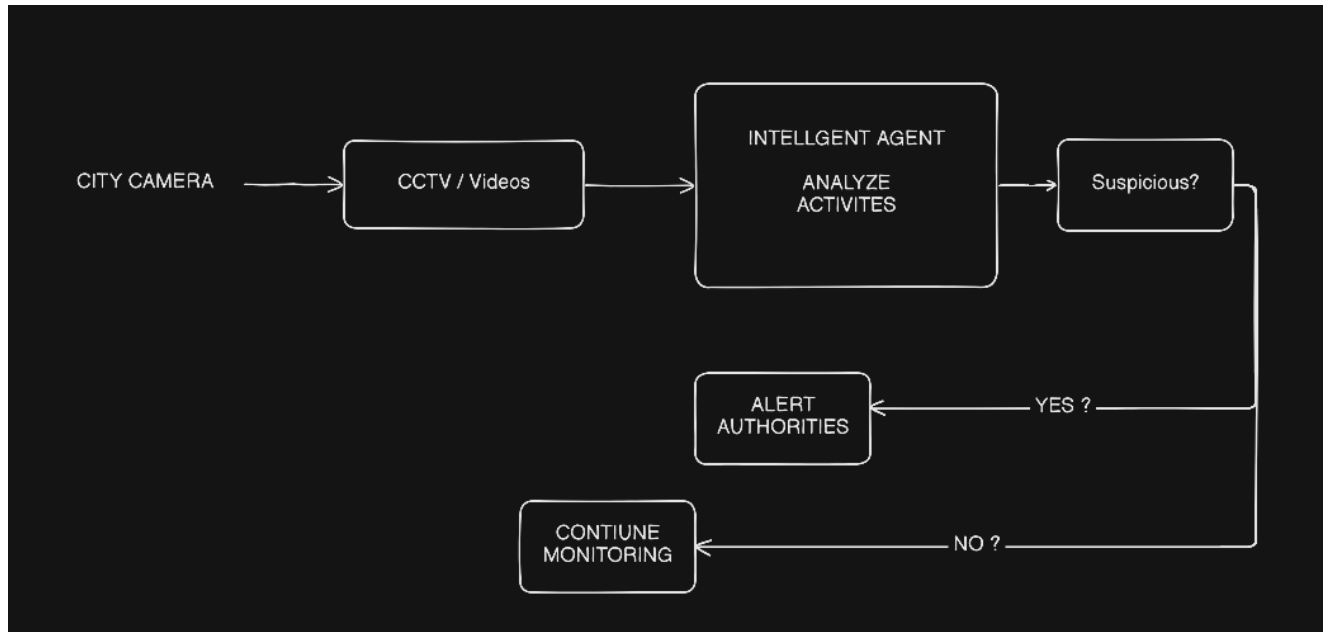
ARTIFICIAL INTELLIGENCE

Assignment no 1

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TITLE:
Smart City Surveillance Agent

Graphical Representation



Textual Representation

1. Sensor: Collects live video data from city surveillance cameras.
2. Actuator: Sends an alert when suspicious activity such as theft, fighting, or unusual behavior is detected.
3. Environment: Streets, markets, roads, public areas, people, and vehicles monitored through city cameras.

PEAS Framework

Component	Description
P – Performance	Accurate suspicious-activity detection
E – Environment	City streets and public areas
A – Actuators	Alerts / notifications
S – Sensors	CCTV / city cameras

Environment Type

Property	Type
Observability	Partially Observable
Determinism	Stochastic
Episodes	Sequential
Change	Dynamic
Values	Continuous
Agents	Multi-Agent

Environment Type Reasoning

Property	Type	One-line Reason
Observability	Partially Observable	Cameras cannot see every area or every action clearly at all times.
Determinism	Stochastic	Human behavior is unpredictable, so the same situation can lead to different outcomes.
Episodes	Sequential	One activity can affect what happens next, so previous actions matter.
Change	Dynamic	People, vehicles, and situations keep changing while the agent is monitoring.
Values	Continuous	Video, movement, speed, and time change continuously rather than in fixed steps.
Agents	Multi-Agent	The environment contains many people and vehicles acting independently.